
PH102: Physics-2

Electrodynamics

Lecture # 13

Electrostatics – Electric fields in
matter – Forces on Dielectrics
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D. Pamu and A. K. Sharma

Previous lecture

- **D** is related to the vacuum field by the relation, $\mathbf{D} = \epsilon_0 \mathbf{E}_{vac}$

- **E** in a medium is related to the vacuum field $\mathbf{E}_{vac}$ by the relation,

$$\mathbf{E} = \frac{1}{\epsilon_r} \mathbf{E}_{vac}$$

- Boundary conditions in terms of free charge on a medium are

$$\epsilon_{above} E_{above}^{\perp} - \epsilon_{below} E_{below}^{\perp} = \sigma_f$$

$$\epsilon_{above} \frac{\partial V_{above}}{\partial n} - \epsilon_{below} \frac{\partial V_{below}}{\partial n} = -\sigma_f$$

$$V_{above} = V_{below}$$

- The energy of the charge configuration in a medium is

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

Interaction potential energy

If the potential is zero at infinity, the energy of a point charge Q is

$$W = QV(\mathbf{r}).$$

For a physical dipole, with $-q$ at $\mathbf{r}$ and $+q$ at $\mathbf{r}+\mathbf{d}$,

$$\begin{aligned} U &= qV(\mathbf{r} + \mathbf{d}) - qV(\mathbf{r}) = q[V(\mathbf{r} + \mathbf{d}) - V(\mathbf{r})] \\ &= q \left[- \int_{\mathbf{r}}^{\mathbf{r}+\mathbf{d}} \mathbf{E} \cdot d\mathbf{l} \right] = -q\mathbf{E} \cdot \mathbf{d} = -q\mathbf{d} \cdot \mathbf{E} \end{aligned}$$

For an ideal dipole the integral reduces to,

$$U = -q\mathbf{E} \cdot \mathbf{d} = -q\mathbf{d} \cdot \mathbf{E}$$

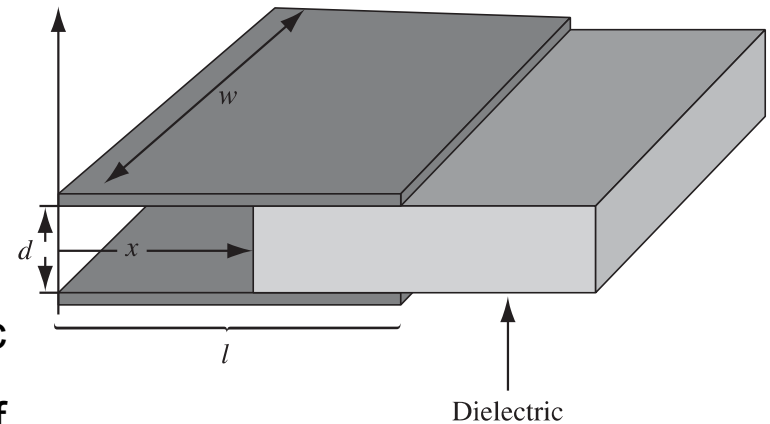
$$U = -\mathbf{p} \cdot \mathbf{E}$$

Forces on dielectrics

Why should there be a force on a dielectric?

The bound charge tends to accumulate near the free charge of the opposite sign and hence experiences an attractive force.

Consider the case of a slab of linear dielectric material, partially inserted between the plates of a parallel-plate capacitor.



We assumed that the field is always uniform inside a parallel-plate capacitor, and zero outside.

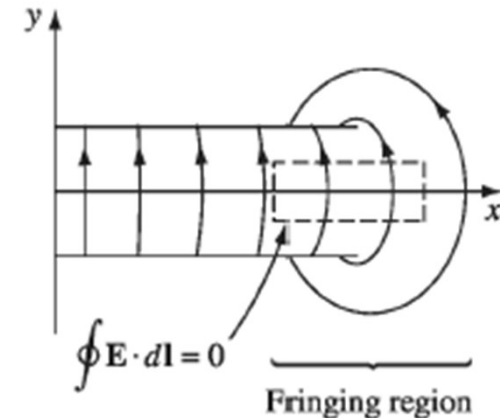
If this is so, then what is the origin of the force?

There is no net horizontal force on the dielectric at all, since the field everywhere would be perpendicular to the plates.

(Contd...)

There is in reality a **fringing field** around the edges, which at least **cannot be ignored this case**. It is this non uniform fringing field that pulls the dielectric into the capacitor.

The field cannot not terminate abruptly at the edge of the capacitor, if it did then $\oint \mathbf{E} \cdot d\mathbf{l} \neq 0$.



If the dielectric is pulled out an infinitesimal distance dx , the energy is changed by an amount equal to the work done,

$$dW = F_{me} \cdot dx$$

F_{me} is the force we must exert, to counteract the **electrical force F** on the dielectric: $F_{me} = -F$. Therefore,

$$F = -\frac{dW}{dx}$$

The energy stored in the capacitor is

$$W = \frac{1}{2} CV^2$$

where,

$$C = C_1 + C_2 = \frac{\epsilon_0 wx}{d} + \frac{\epsilon_0 w(l-x)\epsilon_r}{d} = \frac{\epsilon_0 wx(1-\epsilon_r)}{d} + \frac{\epsilon_0 wl\epsilon_r}{d} = \frac{\epsilon_0 w}{d} (\epsilon_r l - \chi_e x)$$

and l is the length of the plates.

If total charge on the plates ($Q = CV$) is held constant, as the dielectric is moved, then the energy stored in terms of Q is

$$W = \frac{1}{2} \frac{Q^2}{C},$$

therefore,

$$F = -\frac{dW}{dx} = \frac{1}{2} \frac{Q^2}{C^2} \frac{dC}{dx} = \frac{1}{2} V^2 \frac{dC}{dx}$$

Now,

$$\frac{dC}{dx} = -\frac{\epsilon_0 \chi_e w}{d},$$

$$F = -\frac{\epsilon_0 \chi_e W}{2d} V^2$$

The minus sign indicates that the force is in the negative x direction; the dielectric is pulled into the capacitor.

To maintain the capacitor at a fixed potential, by connecting it up to a battery, the battery also does work as the dielectric moves; therefore we now have

$$dW = F_{me} dx + V dQ$$

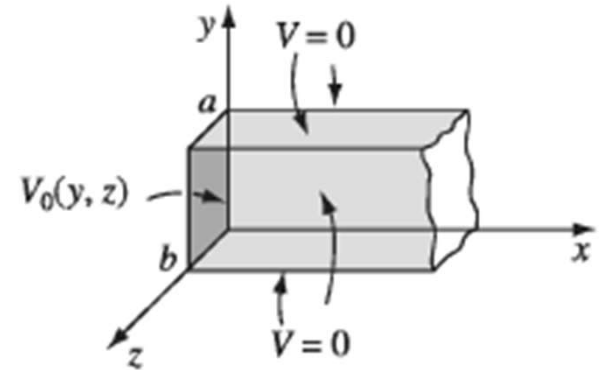
$V dQ$ is the work done by the battery.

$$F = -\frac{dW}{dx} + V \frac{dQ}{dx} = -\frac{1}{2} V^2 \frac{dC}{dx} + V^2 \frac{dC}{dx} = \frac{1}{2} V^2 \frac{dC}{dx}$$

- The force is determined without knowing anything about the fringing fields that are ultimately responsible for it.
- It's built into the whole structure of electrostatics that $\nabla \times \mathbf{E} = 0$, and hence that the fringing fields must be present.

Solved example 1

An infinitely long rectangular metal pipe (sides a and b) is grounded, but one end, at $x = 0$, is maintained at a specified potential $V_0(y, z)$. Find the potential inside the pipe.



➤ 3D problem, therefore

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0,$$

➤ Boundary conditions are

- (i) $V = 0$ when $y = 0$,
- (ii) $V = 0$ when $y = a$,
- (iii) $V = 0$ when $z = 0$,
- (iv) $V = 0$ when $z = b$,
- (v) $V \rightarrow 0$ as $x \rightarrow \infty$,
- (vi) $V = V_0(y, z)$ when $x = 0$.

➤ Solution of the form,

$$V(x, y, z) = X(x)Y(y)Z(z).$$

- Using the solution in Laplace's equation, we get

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0.$$

- The first term depends only on x , the second only on y , and third only on z .
- The sum of these three functions is zero, which implies these three functions must be constant.

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1, \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2, \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = C_3, \quad \text{with } C_1 + C_2 + C_3 = 0.$$

and we now choose C_2 and C_3 such that,

$$\frac{d^2 X}{dx^2} = (k^2 + l^2)X, \quad \frac{d^2 Y}{dy^2} = -k^2 Y, \quad \frac{d^2 Z}{dz^2} = -l^2 Z.$$

- The corresponding solutions can be written as

$$X(x) = A e^{\sqrt{k^2 + l^2} x} + B e^{-\sqrt{k^2 + l^2} x}$$

$$Y(y) = C \sin ky + D \cos ky,$$

$$Z(z) = E \sin lz + F \cos lz.$$

➤ Let us now apply the boundary conditions to evaluate A , B , C , D , E and F .

(v) $\Rightarrow A = 0$.

$$V(x, y, z) = e^{-\sqrt{k^2 + l^2}x} (C \sin ky + D \cos ky) (E \sin lz + F \cos lz)$$

(Absorb B in C and D).

(i) $\Rightarrow D = 0$.

$$V(x, y, z) = e^{-\sqrt{k^2 + l^2}x} (\sin ky) (E \sin lz + F \cos lz)$$

(Absorb C in E and F).

(iii) $\Rightarrow F = 0$.

$$V(x, y, z) = e^{-\sqrt{k^2 + l^2}x} (\sin ky) (E \sin lz)$$

(ii) $\Rightarrow \sin ka = 0$

$$k = \frac{n\pi}{a}, \quad n = 1, 2, 3, \dots$$

(iv) $\Rightarrow \sin la = 0$

$$l = \frac{m\pi}{b}, \quad m = 1, 2, 3, \dots$$

- The final solution is of the form,

$$V(x, y, z) = C e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x} \left(\sin \frac{n\pi}{a} y\right) \left(\sin \frac{m\pi}{b} z\right) \quad (E \text{ replaced by } C)$$

- This solution meets all the boundary conditions except (vi).
- It contains *two* unspecified integers (n and m), and the most general linear combination is a *double* sum:

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x} \left(\sin \frac{n\pi}{a} y\right) \left(\sin \frac{m\pi}{b} z\right)$$

We hope to fit the remaining boundary condition such that

$$V(0, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \left(\sin \frac{n\pi}{a} y\right) \left(\sin \frac{m\pi}{b} z\right) = V_0(y, z)$$

Use the so-called “Fourier’s trick” to find out the coefficients $C_{n,m}$. Multiply the above equation by $\sin(n'\pi y/a)\sin(m'\pi z/b)$ (where n' and m' are positive integers), and integrate the corresponding integrals from 0 to a and 0 to b , respectively,

$$\begin{aligned} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy \int_0^b \sin(m\pi z/b) \sin(m'\pi z/b) dz \\ = \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz. \end{aligned}$$

Using the result of lecture 09, slide 20, we can write

$$C_{n,m} = \frac{4}{ab} \int_0^a \int_0^b V_0(y, z) \sin(n\pi y/a) \sin(m\pi z/b) dy dz.$$

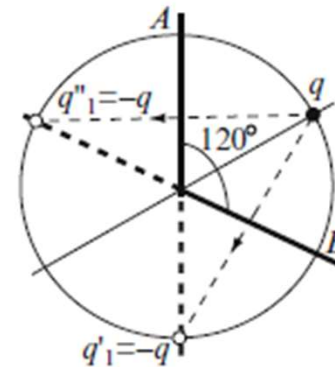
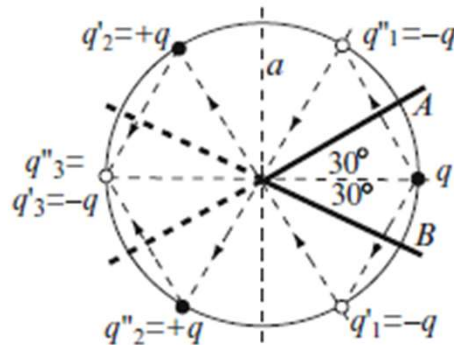
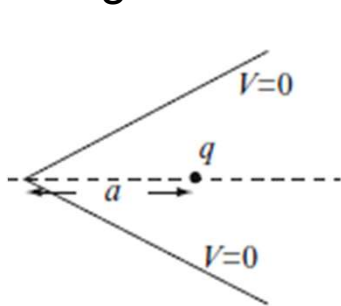
Suppose the strip at $x = 0$ is a metal plate with constant potential V_0 , then

$$\begin{aligned} C_{n,m} &= \frac{4V_0}{ab} \int_0^a \sin(n\pi y/a) dy \int_0^b \sin(m\pi z/b) dz \\ &= \begin{cases} 0, & \text{if } n \text{ or } m \text{ is even,} \\ \frac{16V_0}{\pi^2 nm}, & \text{if } n \text{ and } m \text{ are odd.} \end{cases} \end{aligned}$$

$$\text{Therefore, } V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{n,m=1,3,5\dots}^{\infty} \frac{1}{nm} e^{-\pi\sqrt{(n/a)^2+(m/b)^2}x} \sin(n\pi y/a) \sin(m\pi z/b).$$

Images for two surfaces at any angle

- An image charge is created for any charge reflecting in a conducting surface. Multiple surfaces generate multiple charges.
- In the case of two surfaces, an infinite number of charges are always generated.
- The number of charges is finite only when multiple images happen to fall on each other.
- If multiple reflections at the same location are of the same sign, the problem has an image solution. If not, the problem cannot be solved using the method of images.



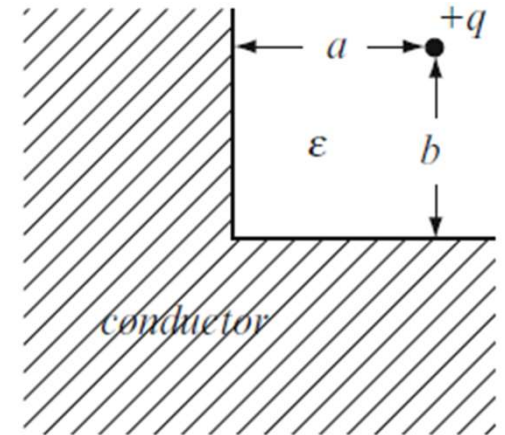
$$N = 2\left(\frac{180^\circ}{\theta}\right) - 1$$

For any integer divisor of 180, n , there are $2n - 1$ images, for a total of $2n$ charges necessary to produce zero potential on the two planes. θ is an integer divisor of 180° .

Solved example 2

A point charge q is located in front of an infinitely long, right-angle conductor in cross section.

(a) Calculate the electric field intensity everywhere in the plane of the charge ($z = 0$).

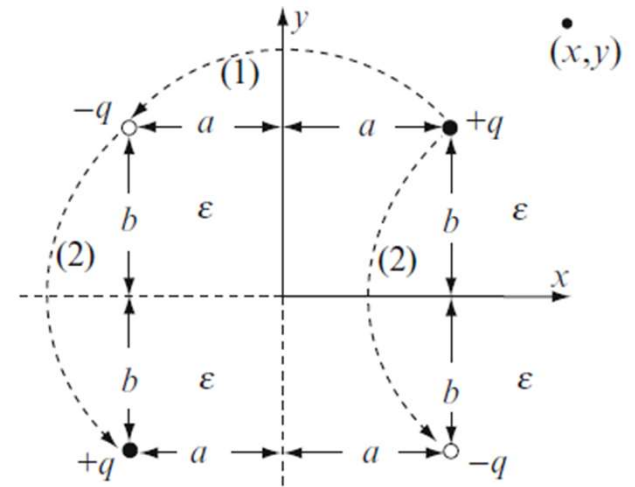


The number of images is given by

$$N = \frac{360^\circ}{\theta} - 1 = \frac{360^\circ}{90^\circ} - 1 = 3$$

The potential due to the four charges is

$$V(x, y) = \frac{q}{4\pi\epsilon} \left[\frac{1}{\left((x-a)^2 + (y-b)^2 \right)^{1/2}} - \frac{1}{\left((x-a)^2 + (y+b)^2 \right)^{1/2}} \right. \\ \left. + \frac{1}{\left((x+a)^2 + (y+b)^2 \right)^{1/2}} - \frac{1}{\left((x+a)^2 + (y-b)^2 \right)^{1/2}} \right]$$



The electric field is

$$\begin{aligned}\mathbf{E}(x, y) &= -\hat{\mathbf{x}} \frac{\partial V(x, y)}{\partial x} - \hat{\mathbf{y}} \frac{\partial V(x, y)}{\partial y} \\ &= \hat{\mathbf{x}} \frac{q}{4\pi\epsilon} \left[\frac{x-a}{\left((x-a)^2 + (y-b)^2\right)^{3/2}} - \frac{x-a}{\left((x-a)^2 + (y+b)^2\right)^{3/2}} + \frac{x+a}{\left((x+a)^2 + (y+b)^2\right)^{3/2}} - \frac{x+a}{\left((x+a)^2 + (y-b)^2\right)^{3/2}} \right] \\ &\quad + \hat{\mathbf{y}} \frac{q}{4\pi\epsilon} \left[\frac{y-b}{\left((x-a)^2 + (y-b)^2\right)^{3/2}} - \frac{y+b}{\left((x-a)^2 + (y+b)^2\right)^{3/2}} + \frac{y+b}{\left((x+a)^2 + (y+b)^2\right)^{3/2}} - \frac{y-b}{\left((x+a)^2 + (y-b)^2\right)^{3/2}} \right]\end{aligned}$$

(b) The charge density induced on the surfaces of the conductor in the same plane as in part (a).

Set $x = 0$ in the expression of the electric field for the charge density on the vertical surface. Therefore,

$$\mathbf{E}(0, y) = -\hat{\mathbf{x}} \frac{qa}{2\pi\epsilon} \left[\frac{1}{\left(a^2 + (y-b)^2\right)^{3/2}} - \frac{1}{\left(a^2 + (y+b)^2\right)^{3/2}} \right]$$

Note that the y component of the field cancels.

(Contd...)

The charge density on this surface is

$$\rho_s(0, y) = \epsilon E(0, y) = -\frac{qa}{2\pi} \left[\frac{1}{(a^2 + (y - b)^2)^{3/2}} - \frac{1}{(a^2 + (y + b)^2)^{3/2}} \right]$$

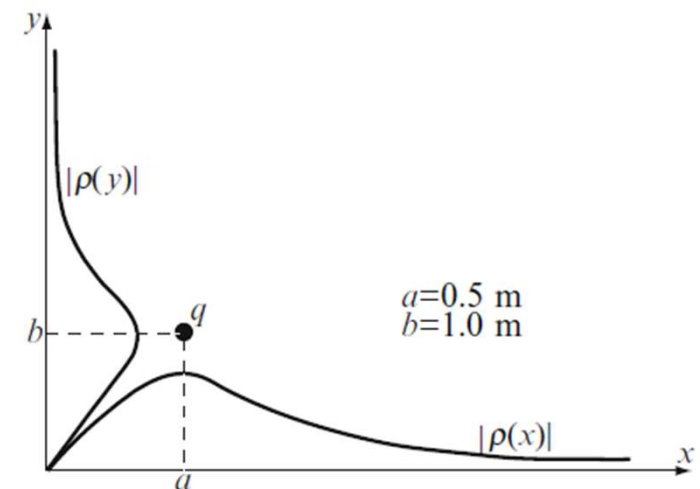
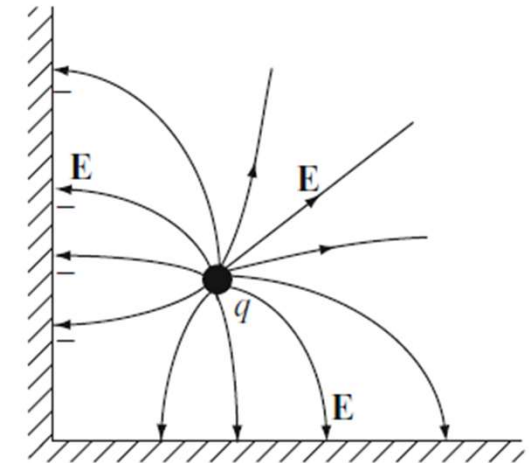
Set $y = 0$ in the expression of the electric field for the charge density on the horizontal surface. Therefore,

$$\mathbf{E}(x, 0) = -\hat{y} \frac{qb}{2\pi\epsilon} \left[\frac{1}{((x - a)^2 + b^2)^{3/2}} - \frac{1}{((x + a)^2 + b^2)^{3/2}} \right]$$

The charge density on this surface is

$$\rho_s(x, 0) = -\frac{qb}{2\pi} \left[\frac{1}{((x - a)^2 + b^2)^{3/2}} - \frac{1}{((x + a)^2 + b^2)^{3/2}} \right]$$

There are two different surfaces with different charge densities.



Questions?